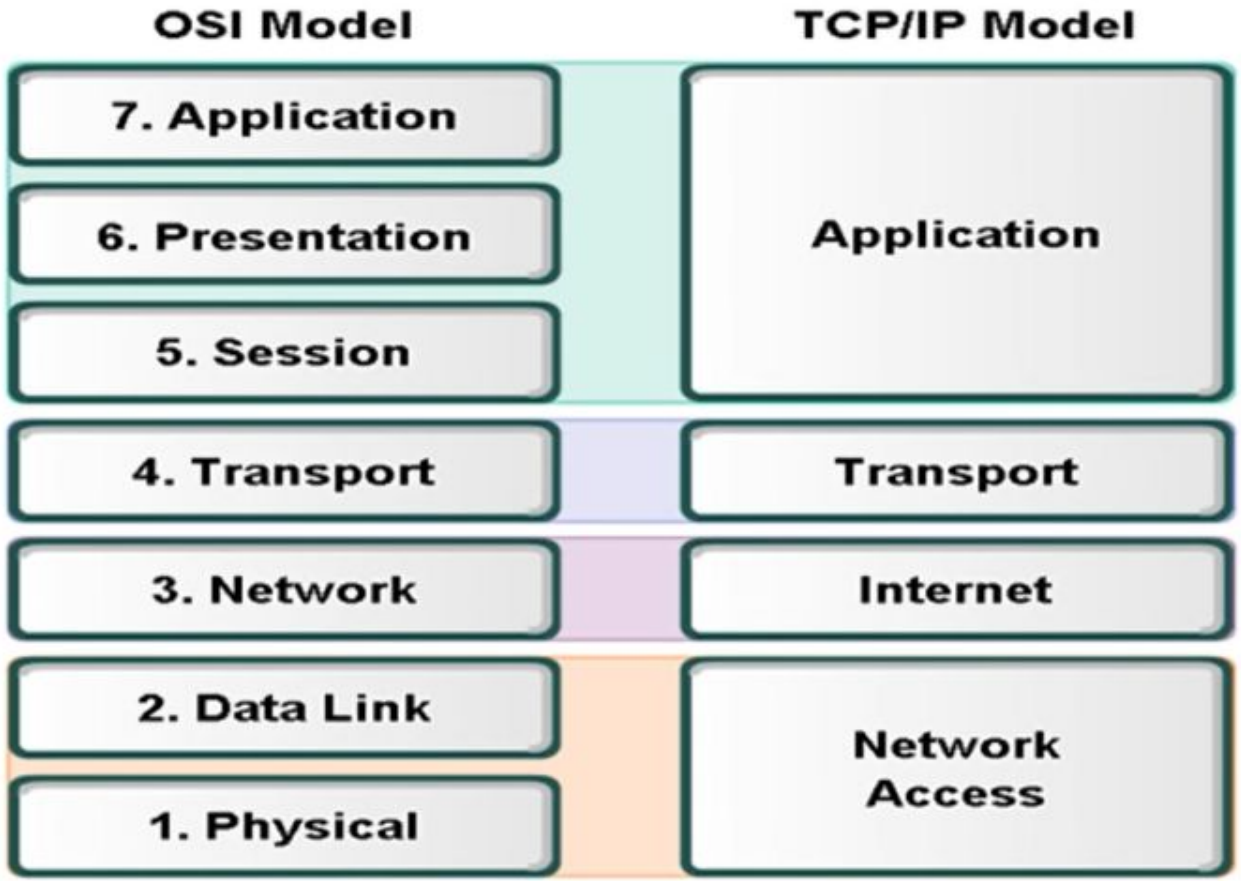
OSI vs. TCP/IP Model

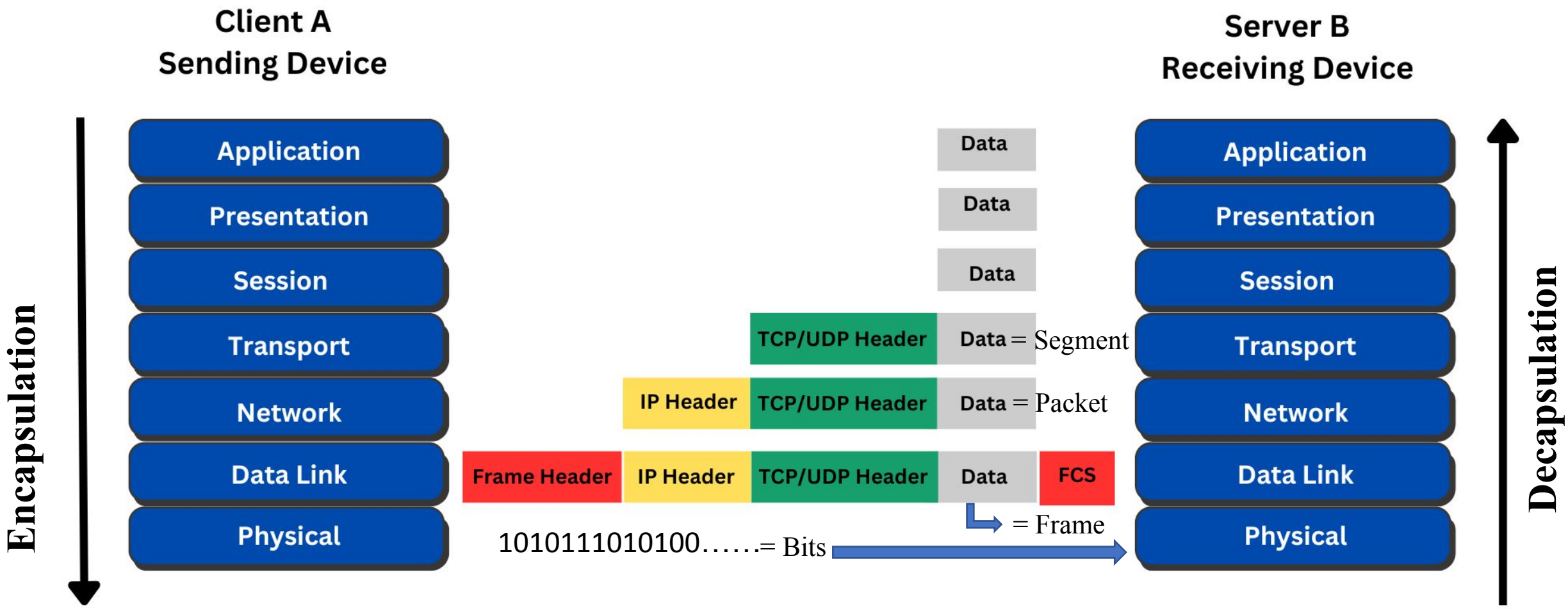


- In the OSI model, there are seven layers, but in the TCP/IP model, only four layers are used.
- The OSI model is a reference model.

Encapsulation/ Decapsulation

- In data communication there are two concept involves in transmission namely encapsulation and decapsulation.
- **Encapsulation** adds addition information to a send packet as it travels to its destination.
- **Decapsulation** is the reverse process of encapsulation, So a receiver node can read the original send data/information.

Encapsulation/ Decapsulation



Encapsulation/ Decapsulation

- **Layers 7-5: Application → Presentation → Session**

 Your data gets formatted, compressed/encrypted, and tagged with a session ID

(Like writing your letter and putting it in an envelope)

- **Layer 4: Transport**

 Data split into **segments** (TCP) or **datagrams** (UDP)

 Adds: source/destination ports + sequence numbers

(Like breaking a large shipment into boxes and numbering them)

- **Layer 3: Network**

 Segment becomes a **packet**

 Adds: source/destination IP addresses

(Like adding sender/receiver addresses on the box)

Encapsulation/ Decapsulation

- **Layer 2: Data Link**

- The packet flows down to the Data Link layer, where a new header containing the source MAC address and destination MAC address is added.
- A trailer is also added at the Data Link Layer called the Frame Check Sequence (FCS). This is known as a frame.

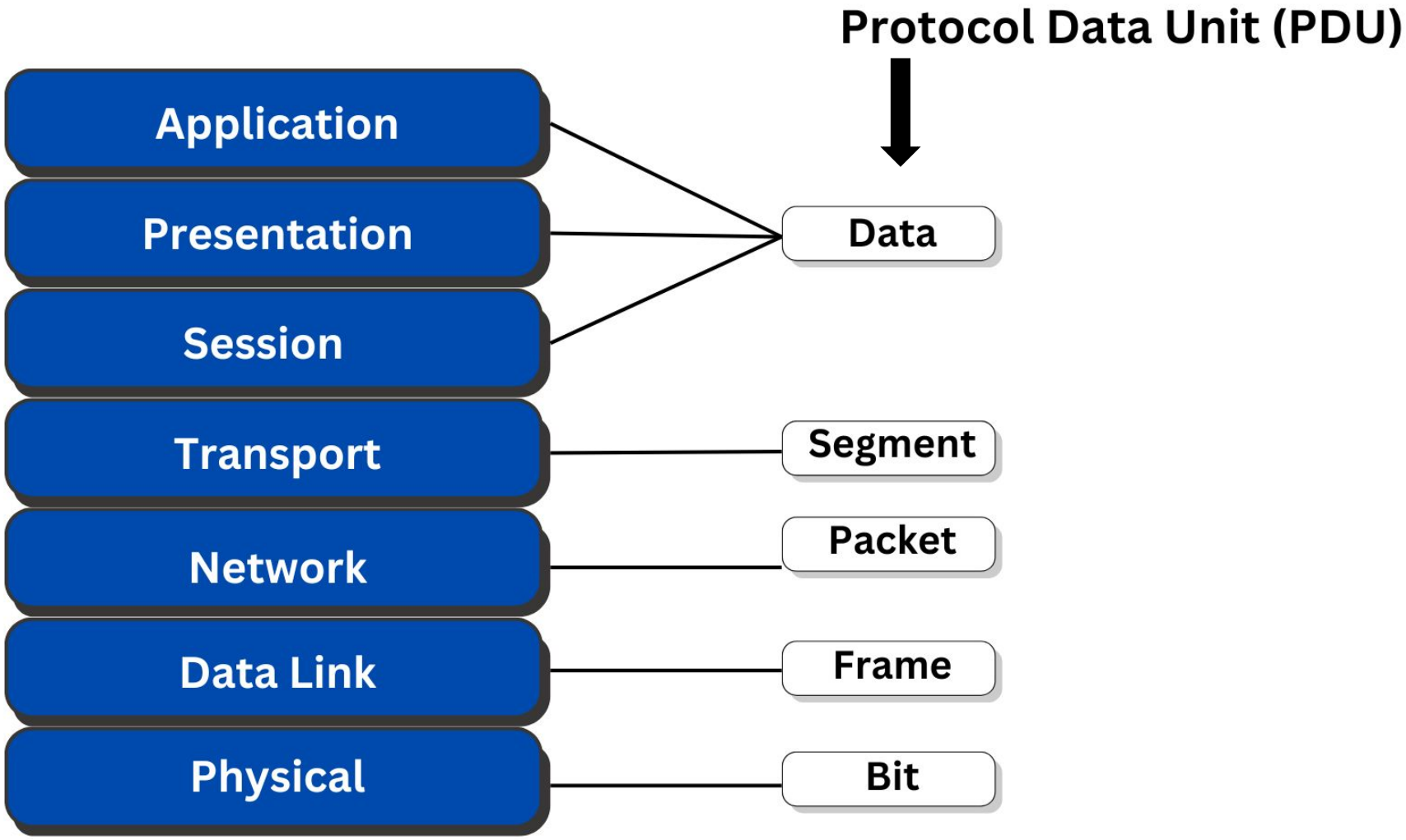
(Like putting the box on a delivery truck with a tracking barcode)

- **Layer 1: Physical**

- Frame becomes electrical/light **signals**

(The actual delivery through cables or air)

Encapsulation/ Decapsulation



In **Encapsulation** Process, PDU sequence: Data ⟶ Segment ⟶ Packet ⟶ Frame ⟶ Bits

In **Decapsulation** Process, PDU sequence: Bits ⟶ Frame ⟶ Packet ⟶ Segment ⟶ Data

PDU

Each layer's information is referred to as a Protocol Data Unit (PDU). Along with the data, it contains protocol-specific control information.

Each layer will add (or delete) its protocol information as a PDU moves down (or up) each layer.

Layer	**PDU Name**	**PCI Contains**
Layer 4 (Transport)	Segment/Datagram	Port numbers, sequence numbers, flow control
Layer 3 (Network)	Packet	Source/destination IP addresses, TTL
Layer 2 (Data Link)	Frame	Source/destination MAC addresses, FCS
Layer 1 (Physical)	Bit/Symbol	N/A (physical encoding)

What's the PDU at Layer 3?

→ Packet